Furuta Pendulum Model Decision Guide

Substantive differences only

The common comparison uses l1 = lc, l0 = J0, and J1 = J1r only when the axes, reference points, and included hardware are the same. With s = sin(theta1), c = cos(theta1), and delta = m1 lc^2 + J1t - J1a, the exact missing mechanical content is summarized by the following equations.

$$H(\theta_1) = J_{1a}\cos^2(\theta_1) + (m_1 l_c^2 + J_{1t})\sin^2(\theta_1)$$

$$T_A - T_B = \tfrac{1}{2} H(\theta_1)\dot{\theta}_0^2$$

$$M_A - M_B = \left[[H(\theta_1), 0], [0, 0]\right]$$

$$C_A - C_B = \left[2\delta \sin(\theta_1)\cos(\theta_1)\,\dot{\theta}_0\dot{\theta}_1;\ -\delta \sin(\theta_1)\cos(\theta_1)\,\dot{\theta}_0^2\right]$$

Plain-English meaning. The MATLAB model removes part of the arm-axis kinetic energy. It therefore uses a constant arm-axis inertia and loses the two nonlinear velocity forces created when the full inertia changes with pendulum angle.

Real difference	PDF model	MATLAB model	Required action	Condition for a reduced alternative
Three-dimensional COM motion	Includes radial, tangential, and vertical COM velocity. The radial term is -lc sin(theta1) theta0dot.	Uses vx and vy only. The radial velocity is absent.	Keep the PDF COM position and full velocity in the canonical plant. Do not substitute the two-component tutorial velocity.	Allow the planar velocity only for a stated small-angle, low-speed study after numerical comparison over the intended envelope meets a tolerance chosen before testing.
Pendulum rotational energy	Retains hinge, transverse, and longitudinal COM principal inertias J1r, J1t, and J1a.	Retains only J1 theta1dot^2/2, naturally mapped to hinge-axis inertia J1r.	Obtain all three COM inertias from CAD or identification and retain them. Record axes and reference point.	A single inertia is acceptable only if the omitted arm-induced rotational energy is shown negligible over the operating envelope.
Arm-axis mass-matrix entry	M00 = A(theta1), so arm-axis inertia changes with pendulum angle.	M00 = I0 + m1 L0^2, a constant.	Keep A(theta1). Use the exact full-model expression in MATLAB and Simulink.	Near upright, the linearized matrices can match if I0 = J0 + J1a and J1 = J1r. This does not establish nonlinear equivalence.
Mixed arm velocity force	Includes 2 delta sin(theta1) cos(theta1) theta0dot theta1dot.	Term is absent.	Keep the term in the canonical nonlinear equation.	Omit only in a separately named reduced model whose acceleration and trajectory errors pass the agreed envelope test.
Pendulum centrifugal force	Includes -delta sin(theta1) cos(theta1) theta0dot^2.	Entire term is absent.	Keep the term, especially for swing-up and fast arm motion.	Omit only when arm speed is limited and validation proves the term has no material effect on the intended controller or trajectory.
Plant input boundary	Validated mechanical plant input is effective arm-axis torque tau.	Generated plant input is voltage after algebraic current elimination; mechanicalEqs still expose torque.	Maintain separate named interfaces for torque input and voltage input. Connect them through an explicit actuator subsystem.	Compare models directly only after voltage, current, gear ratio, back-EMF, and torque are mapped to the same arm-axis torque.
Electrical state order	Proposes an inductive motor extension with current as a fifth state.	Sets motor inductance to zero, so current is algebraic and the plant has four states.	Implement the five-state model as the reference actuator. Derive the four-state version from it.	Use the four-state actuator only if Lm/Rm is small relative to the fastest relevant mechanical and control time scales and a response comparison passes.
Transmission and rotor inertia	Extension defines N, efficiency, and reflected rotor inertia N^2 Jm, with a warning to count it once.	Assumes direct drive N = 1, ideal transmission, and no explicit reflected rotor inertia.	Define the actual speed ratio and sign. Use J0eff = J0mech + N^2 Jm exactly once. Start with an ideal transmission or a measured loss model.	Use N = 1 only for genuine direct drive. Use a gearbox or belt model when hardware changes motor speed or torque at the arm.
CAD inertia handling	Uses a documented body frame with three principal inertias and states that products of inertia need an extension.	Provides no full inertia-tensor or CAD-frame mapping.	Rotate each CAD tensor into the adopted model frame. Retain products of inertia if the selected axes are not principal.	Use a diagonal tensor only when the CAD frame is principal or the off-diagonal terms are shown negligible for the assembly.
Validation and parameter protection	Includes symbolic, dimensional, energy, equilibrium, and mass-matrix checks with parameter provenance.	Symbolic template has no parameter-domain checks or built-in numerical regression tests.	Add named parameters with units and provenance, domain checks, and automated model tests before team use.	A bare symbolic template is acceptable only for private exploration, not as the shared project baseline or controller plant.

Deliberately excluded from the table. J0 versus I0, J1r versus J1, and lc versus I1 are only notation changes after physical axes and reference points are confirmed. The two state orders differ by a permutation. The

potential energies differ by a constant. The tangential coupling, hinge inertia, gravity torque, and viscous damping are otherwise the same under matched parameters. None of these is a physical model difference.

What to add to the PDF model

1. Add a motor and transmission layer without changing the mechanical mass matrix or nonlinear force vectors.
2. Use one documented arm-side torque convention. Define motor speed by omega_m = N theta0dot, including the sign of N.
3. Include motor resistance, inductance, torque constant, back-EMF constant, and rotor inertia using measured or manufacturer data. Include reflected rotor inertia once.
4. Add named parameter records with units, source, reference point, axis, and inclusion notes for each inertia.
5. Add CAD tensor rotation, parameter-domain checks, mass-matrix conditioning checks, energy and equilibrium tests, and side-by-side comparison with Simscape Multibody.
6. Add saturation, dead zone, Coulomb friction, backlash, compliance, or sensor dynamics only when measurements justify those terms.

Recommended unified equations

Keep the PDF mechanical dynamics and supply arm-axis torque from the actuator:

$$M_{A(q)}\ddot{q} + C_{A(q,\dot{q})} + G_{A(q)} + D_{A(\dot{q})} = [\tau_a;\ 0]$$

For an ideal transmission with omega_m = N theta0dot, use the five-state electrical model:

$$\tau_a = N K_t i$$

$$L_m i = V_m - R_m i - K_e N \dot{\theta}_0$$

Use state x = [theta0, theta1, theta0dot, theta1dot, i]^T and voltage input u = Vm. If measured transmission losses must be included, use a direction-consistent loss or friction model. A single constant efficiency in only one power direction should not be treated as a globally energy-consistent gearbox model.

If the electrical time-scale test supports Lm = 0, derive the reduced actuator rather than writing a separate unrelated formula:

$$i \approx \frac{\left(V_m - K_e N \dot{\theta}_0\right)}{R_m}$$

$$\tau_a \approx \left(N\frac{K_t}{R_m}\right)V_m - \left(\frac{N^2 K_t K_e}{R_m}\right)\dot{\theta}_0$$

Implementation rule. Solve M_A qddot = rhs as a linear system. Do not form the matrix inverse numerically.

What to remove or avoid

Remove from the PDF mechanics. Nothing. The full kinematic, inertia, gravity, and viscous terms should remain in the canonical model. A simplification must be created as a separate reduced model so that the physical baseline is preserved.

- Do not remove the radial COM velocity, the J1t and J1a energy terms, A(theta1), or either delta-dependent nonlinear velocity term merely to make the equations match the MATLAB template.
- Do not count N^2 Jm both inside J0 and again as a separate mass-matrix contribution.
- Do not call torque-input and voltage-input plants equivalent until the actuator equations have reconciled their inputs.

- Do not replace the pivot-to-COM distance lc with total pendulum length unless the COM is actually at that location.
- Do not use the MATLAB state permutation silently at interfaces. Name the state order in every function, bus, and logged dataset.
- Do not add friction nonlinearities, saturation, efficiency, or sensor dynamics as guessed constants. Keep them outside the baseline until data exist.

Conditions for choosing each implementation

Implementation	Use it when	Do not use it when
Full PDF mechanics plus five-state motor	This is the default when motor inductance is known or may affect current transients, saturation, sampling, or closed-loop design. It also provides the reference against which a reduced actuator is tested.	Do not use an unidentified electrical model. If Lm, Rm, Kt, or Ke are unknown, keep the mechanics torque-driven until those parameters are obtained.
Full PDF mechanics plus four-state motor	Use after tau_e = Lm/Rm is demonstrably short relative to the fastest relevant mechanical mode and control or sampling time, and the five-state comparison stays within the agreed tolerance.	Do not use when current peaks, electrical transients, or current limiting affect performance or safety.
Simplified MATLAB mechanics	Use as a teaching model, a computational benchmark, or a local near-upright model after parameter remapping and envelope validation. Near upright, linearizations match if I0 = J0 + J1a and J1 = J1r.	Do not use as the general swing-up, high-speed, CAD, or Simscape reference. Do not claim exact nonlinear equivalence under the natural parameter meanings.
Another proposed model	Use only after its coordinates, energy, input boundary, parameter sources, and validation evidence satisfy every team acceptance condition below.	Do not adopt it because it is shorter, already coded, visually similar, or produces one plausible trajectory.

Team acceptance conditions

The team should follow this recommendation unless an alternative passes every applicable condition below. The team should set its numerical tolerances before seeing comparison results.

1. Coordinate agreement. The alternative states the inertial frame, hinge axis, zero angles, positive directions, state order, and gravity direction, and reproduces the same physical configuration.
2. Parameter traceability. Every mass, COM location, inertia, damping coefficient, motor constant, gear ratio, and reflected component has units, a reference point, an axis, and a source.
3. Energy derivation. Every retained kinetic-energy term follows from an explicit COM velocity or body angular velocity. Every omitted term is identified, not hidden inside a name change.
4. Input agreement. Torque, current, and voltage models are compared only after transmission ratio, efficiency or losses, back-EMF, and rotor inertia are reconciled at the same arm-side boundary.
5. Structural tests. The mass matrix is symmetric and positive definite for the allowed parameter set. Upright and downward gravity signs are correct. Conservative energy and damped power balance pass.
6. Envelope test. Acceleration, trajectory, energy, and linearization differences are evaluated over the intended angles, speeds, inputs, initial conditions, and controller bandwidth, not at one operating point.
7. Independent reference. The proposed model agrees within the preselected tolerance with the full PDF equations, an aligned Simscape Multibody model, or measured data appropriate to its claimed scope.
8. Controller relevance. Any remaining difference is shown not to change the intended stabilization, swing-up, saturation, or safety conclusions.
9. Reproducibility. Another student can run the same tests from a named parameter set and obtain the same pass or fail result.
10. Scope label. A reduced model is named and documented as reduced. It does not silently replace the canonical full nonlinear model.

Why should this justification govern the team decision

The PDF model earns its role as the mechanical baseline because each extra term can be traced to a physical velocity or angular-velocity component. The exact symbolic difference shows where the models diverge: one additional arm-axis energy contribution H(theta1) and the nonlinear forces generated by it. These terms vanish from a first-order upright linearization under specific parameter mapping, but they do not vanish from general swing-up motion.

The MATLAB model remains useful because its Euler-Lagrange steps, damping forces, and algebraic motor elimination are coherent for the simplified energy it declares. Its motor equations can therefore be reused without accepting its reduced mechanics. This separation gives the team a clear rule: choose mechanical fidelity from the PDF, then choose four or five actuator states from measured electrical time scales.

Exact nonlinear equivalence is not a reasonable default. If I0 may absorb J1a, equality still requires J1a = m1 lc^2 + J1t so that delta = 0. That special identity is not a general property of a Furuta pendulum. Without that identity, or without an error study that supports a reduced range, calling the two nonlinear models equivalent would be mathematically unsupported.

A synthetic check in the comparison record also produced different accelerations away from equilibrium. That result is not a hardware error estimate, but it confirms that the omitted terms change predictions. The defensible project practice is therefore to preserve the full model and make every approximation explicit, testable, and reversible.

Questions the team must answer

- Is the arm genuinely direct drive, or is there a gearbox, belt, or other transmission? What is the signed motor-to-arm speed ratio N?
- What are the measured or manufacturer values of Rm, Lm, Kt, Ke, and motor rotor inertia Jm?
- Is Lm/Rm small compared with the controller sample time and the fastest mechanical or closed-loop time scale?
- What exactly is included in J0: bare arm, hub, fixtures, encoder components, rotor reflection, or other attachments?
- Does l1 in the existing code mean pivot-to-COM distance, and has that COM been obtained from CAD or measurement?
- Can CAD provide mass, COM, full inertia tensor, reference point, and coordinate axes for each assembled moving body?
- Are the pendulum products of inertia negligible after rotation into the selected body frame?
- What angle range, speed range, initial conditions, swing-up trajectory, and controller bandwidth define the required operating envelope?
- What voltage, current, torque, and speed limits does the driver and motor impose?
- Are Coulomb friction, stiction, backlash, compliance, or cable torque visible in experiments, and can they be identified rather than guessed?
- What quantitative tolerances will the team use for acceleration, trajectory, energy, Simscape, and controller comparisons?
- Which model names and state ordering will be fixed for the torque-input, reduced voltage-input, and inductive voltage-input plants?

Final instruction to the project team

Adopt the PDF mechanics unchanged. Add the motor as a modular actuator, begin with the five-state electromechanical formulation when inductance data are available, and approve the four-state reduction only after the time-scale and response tests pass. Preserve the simplified MATLAB mechanics as a labeled comparison model. Choose a different foundation only if it satisfies the full team acceptance conditions with reproducible evidence over the project's intended operating envelope.